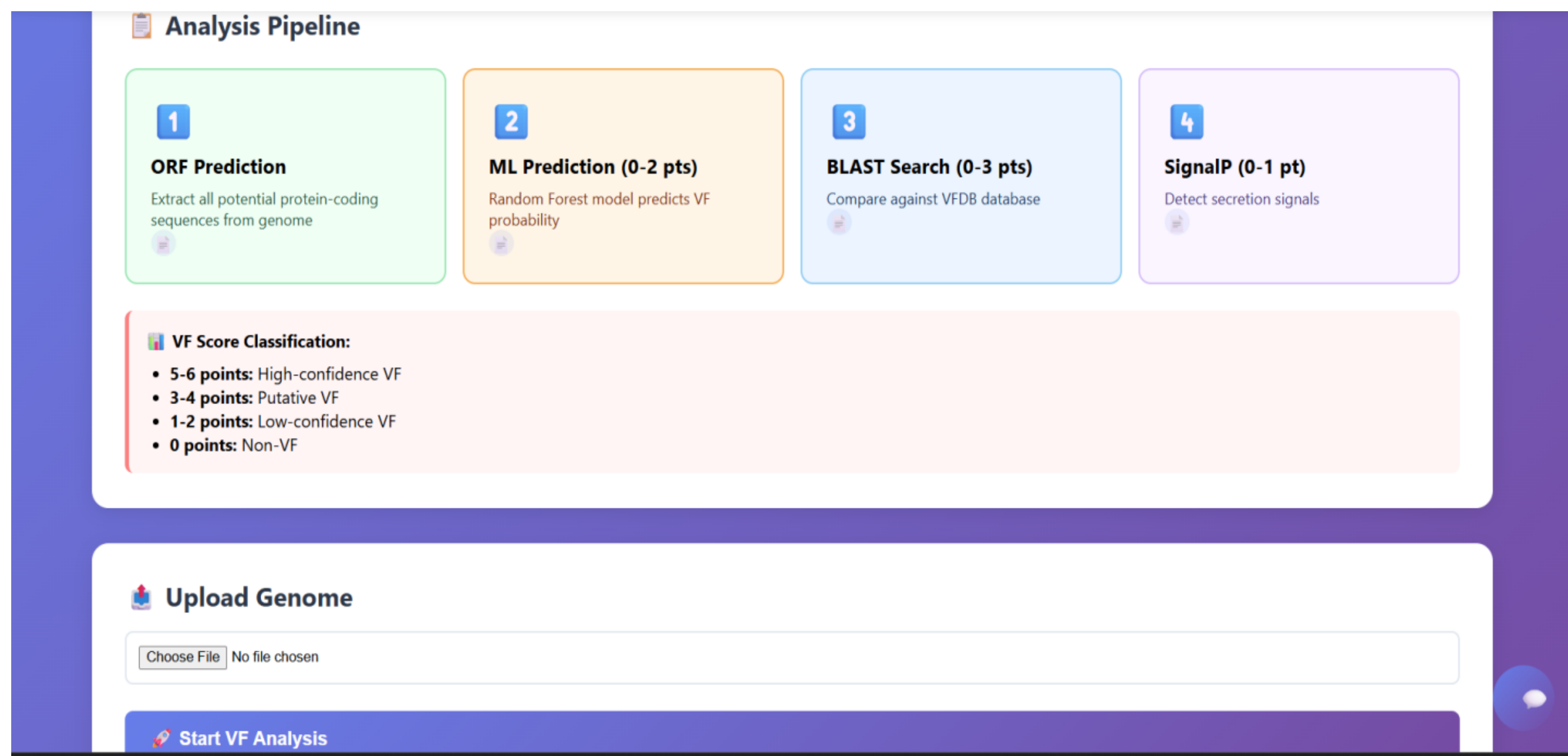


COMPUTATIONAL DETECTION OF VIRULENCE FACTORS IN MICROBIAL GENOMES.

ABSTRACT:

Virulence factors (VFs) allow microbes to infect hosts, making their detection crucial. Experimental methods are time-consuming and hard to scale, so computational approaches are needed. We developed an integrated pipeline using machine learning, database similarity searches, HMM domain analysis, and signal peptide prediction. Results are combined with a confidence-scoring system for reliable interpretation. The pipeline is available as a web platform for interactive analysis, visualization, and bioinformatics education..

METHODS & MATERIALS:



1. Genome Data Preparation A microbial genome was processed to identify ORFs, which were translated into protein sequences.
 2. Machine Learning Prediction Sequence features trained a Random Forest classifier to score proteins for virulence likelihood.
 3. Sequence Similarity (BLAST)BLASTP searches against VFDB identified homologs of known virulence proteins.
 4. Domain-Based Detection HMMER with curated PFAM profiles detected conserved virulence-associated domains.
 5. Signal Peptide Prediction Proteins likely to be secreted were identified to provide biological context.
 6. Evidence Integration and Scoring All results were combined into a confidence-based scoring system for classification.
 7. Web Platform Deployment
- The pipeline was made available as a web app for interactive analysis, visualization, and bioinformatics learning.

References:

Chen L. et al. VFDB: bacterial virulence factor database. Nucleic Acids Res. Altschul S.F. et al. BLAST sequence alignment. Nucleic Acids Res. Eddy S.R. Profile HMM searches (HMMER). PLoS Comput Biol. Petersen T.N. et al. SignalP for signal peptide prediction. Nat Methods. Breiman L. Random Forests. Machine Learning. PathoFact : automated virulence factor prediction pipeline.

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INTRODUCTION:

Virulence factors (toxins, adhesins, secretion systems) are central to microbial pathogenicity. Computational approaches are needed to scale VF detection, but single-method predictions are often limited. Integrating multiple computational strategies improves prediction accuracy and interpretability.

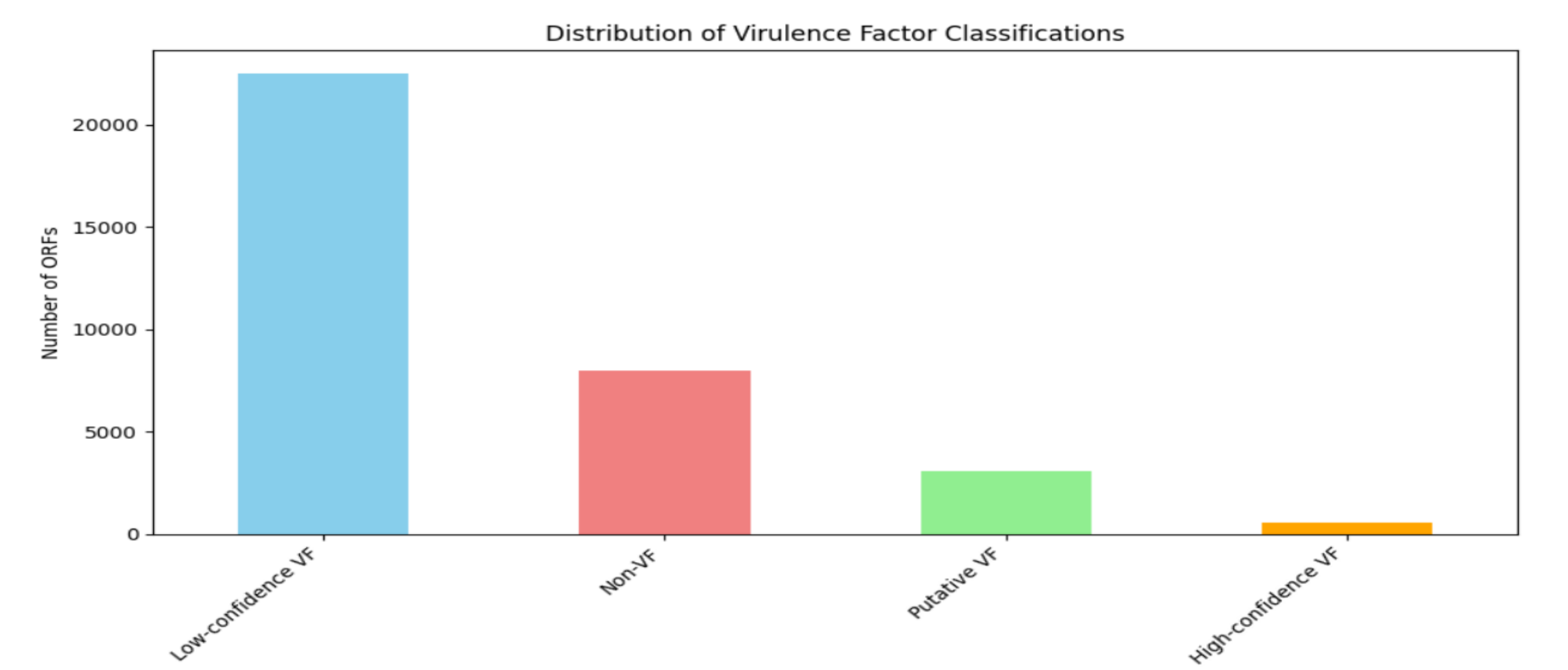
OBJECTIVE:

- Develop an integrated computational pipeline for VF detection.
- Combine multiple biological evidence sources.
- Provide an accessible web-based analysis platform.

KEY FEATURES:

- Machine learning–based VF Prediction.
- BLAST similarity search against curated VF databases.
- Domain-based profiling and signal peptide detection.
- Unified confidence scoring system.

RESULTS:



- **Machine Learning Performance:**
Accuracy 0.817 | Precision 0.820 | Recall 0.832 | F1-score 0.826 | AUC 0.898
- **Virulence Classification (34,074 ORFs):**
 - o High-confidence VFs: 551
 - o Putative VFs: 3,060
 - o Low-confidence VFs: 22,504
 - o Non-VFs: 7,959

DISCUSSION:

Each method captures distinct biological signals: BLAST identifies known homologs, HMM detects conserved virulence domains, ML identifies broader sequence patterns, and signal peptide analysis provides functional context. Their integration improves confidence in predicted VFs while reducing false positives. The web-based platform enhances accessibility, reproducibility, and educational value.

CONCLUSION & FUTURE WORK:

I developed a reproducible pipeline for detecting virulence factors in microbial genomes by integrating machine learning, sequence similarity, domain analysis, and signal peptide prediction. The unified scoring system improves reliability and biological interpretation, while the web platform enhances accessibility and educational value. Future work will include secretion system classification, virulence regulation analysis, and explainable AI (XAI) to improve biological insight and model transparency.

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